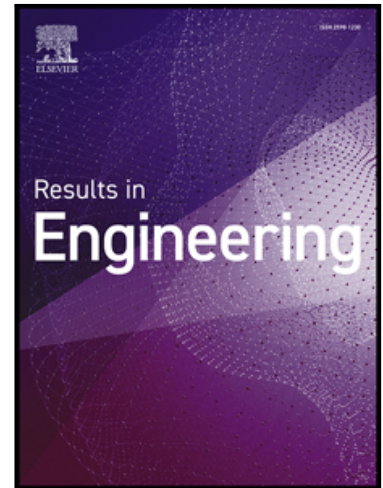


Intelligent Digital Twin for Real-Time Remaining Useful Life
Prediction of Lithium-ion Batteries with Cumulative Incremental
Learning and Ensemble Strategy



Mehran Haghighi , Mohammad Mohammadi , Mehdi Allahbakhshi

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Highlights

- An intelligent digital twin framework is proposed for real-time capacity estimation and RUL prediction of lithium-ion batteries.
- Five deep learning architectures are integrated with cumulative incremental learning that retrains on entire historical data after each update.
- Six ensemble aggregation methods are evaluated, with optimization-based ensembles (NMOE, ERWE) significantly outperforming single models.
- On NASA PCoE dataset, the framework improves R^2 from 0.922 to 0.956 and achieves zero-cycle error in RUL prediction.

Intelligent Digital Twin for Real-Time Remaining Useful Life Prediction of Lithium-ion Batteries with Cumulative Incremental Learning and Ensemble Strategy

Mehran Haghighi, Mohammad Mohammadi, Mehdi Allahbakhshi*

Department of Power and Control Engineering, School of Electrical and Computer Engineering, Shiraz University, Shiraz, Iran

*Corresponding Author: Mehdi Allahbakhshi, Allahbakhshi@shirazu.ac.ir

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ABSTRACT

The gradual capacity degradation of lithium-ion batteries significantly impacts the operational reliability and performance of electric vehicles, and accurate real-time prediction of their remaining useful life continues to be one of the major challenges within battery management systems. In this study, an intelligent digital twin framework is proposed for simultaneous capacity estimation and remaining useful life prediction of lithium-ion batteries, which enables real-time model updating and stable prediction through cumulative incremental learning and ensemble aggregation. The framework integrates five deep learning architectures, namely Stacked LSTM, TCN-LSTM, 1D-CNN-BiLSTM, Transformer Encoder, and Conformer. Each model is initially trained on historical data and subsequently updated upon receiving real-time data through the cumulative incremental learning approach. The outputs of the updated models are combined using six ensemble aggregation strategies, and the overall framework performance is assessed via a three-level evaluation protocol using the NASA Prognostics Center of Excellence and Oxford Battery Degradation datasets. The results confirm that incremental learning improves the performance of all architectures; Conformer, as the best individual model, improved from $R^2 = 0.922$ in offline mode to $R^2 = 0.956$ on the NASA dataset and from $R^2 = 0.884$ to $R^2 = 0.959$ on the Oxford dataset. At the ensemble level, the Nelder-Mead Optimized Ensemble strategy achieved the best performance with $R^2 = 0.963$ and zero-cycle error in remaining useful life prediction on the NASA dataset, while achieving $R^2 = 0.983$ on the Oxford dataset. This framework demonstrates that integrating cumulative incremental learning, architectural diversity, and ensemble aggregation within a digital twin paradigm enables accurate and stable battery state prediction under real-time operational conditions and opens a new pathway toward developing intelligent battery management systems for electric vehicles.

1. Introduction

In recent decades, the energy crisis and environmental pollution have emerged as two major and enduring challenges facing the global community. This situation has turned the active development of the electric vehicle (EV) industry into a strategic objective and a top priority in many countries. In this regard, the European Union has effectively banned the sale of new diesel and gasoline vehicles beyond 2035 by imposing stringent regulations[1]. Lithium-ion batteries, serving as the core component of electric vehicles, have consequently gained significant research interest as well[2]. These batteries are widely employed in electric vehicles owing to their favorable properties, such as rapid charging capability, long cycle life, high energy density, rechargeability, broad operating temperature range, and low self-discharge rate[3, 4]. Nevertheless, one of the primary concerns regarding the lifespan of lithium-ion batteries is the gradual degradation of capacity, which directly affects the reliability and driving range of electric vehicles. When the capacity drops to approximately 70–80% of its initial value, the battery is considered to have reached its end-of-life threshold, and further operation past this point can result in operational failures and even catastrophic incidents[5]. In addition, premature replacement of the battery would lead to an unnecessary waste of battery resources[6, 7]. With the ever-growing adoption of lithium-ion batteries, the need for efficient battery management, particularly real-time monitoring and prediction of the remaining useful life (RUL) of batteries, has become increasingly critical. To ensure safety, reliability, and extended overall longevity of battery systems, the development of battery management systems (BMS) is essential[8]. Accurate real-time prediction of RUL is of critical importance, as it provides timely information on battery aging and can help prevent potential failures[9]. In recent years, numerous research approaches have been developed for estimating battery capacity and predicting RUL. These methods are generally classified into two main categories: model-based methods and data-driven methods. Model-based prediction methods estimate the RUL

of lithium-ion batteries through mathematical equations. However, the implementation of these methods is challenging, as they require in-depth domain expertise for accurate determination of model parameters and demand a substantial volume of experimental data for successful validation[10]. Data-driven methods, on the other hand, carry out RUL prediction based on the analysis of operational data without requiring physical knowledge; nevertheless, the accuracy of these methods depends on the availability of a sufficient volume of historical data. The electrochemical model (EM), equivalent circuit model (ECM), and empirical model are frequently used for capacity estimation or RUL prediction of lithium-ion batteries within model-based approaches. Zheng et al.[11] employed a pseudo-two-dimensional (P2D) electrochemical model integrated with proportional-integral observers for battery capacity estimation; however, the performance of this model deteriorates noticeably at low charge and discharge rates. Sadabadi et al.[12] presented a framework utilizing electric vehicle charging data for RUL prediction through optimizing the parameters of a single-particle model; however, the inherently dynamic and nonlinear behavior of electrochemical reactions ultimately limited the accuracy of the model. In ref[13], a simplified electrochemical model was developed by simultaneously incorporating capacity and voltage data, and its robustness to noisy data was confirmed. Furthermore, Gao et al.[14] modeled the battery life degradation pattern based on electrochemical principles; although the prediction accuracy was improved, the determination of model parameters remained a persistent challenge. Overall, while electrochemical models provide a fundamental understanding of battery aging mechanisms, their high computational complexity and the difficulty associated with parameter estimation pose significant obstacles to their practical applicability. Guha et al.[7] developed an algorithm based on a fractional-order equivalent circuit model with real-time execution capability for RUL prediction; however, the complexity of the model and the limitations of laboratory conditions reduced its generalizability. Shi et al.[15] estimated battery capacity

by constructing a second-order resistance-capacitance (RC) model and utilizing a hybrid of the unscented particle filter (UPF) and the unscented Kalman filter (UKF). Generally, equivalent circuit models reformulate the RUL prediction problem into a parameter estimation task and depend on filtering algorithms, including the extended Kalman filter (EKF) and particle filter (PF). Nonetheless, the strong dependence on operating conditions and the difficulty of generalization to real-world scenarios remain the primary challenges associated with these methods. Empirical models represent the battery degradation trend using simple mathematical functions. For instance, Tian et al.[16] introduced a semi-empirical model for RUL prediction, while in ref[17], a two-segment logarithmic model was developed. Although these models are simpler than electrochemical approaches, they exhibit limited generalizability. Among data-driven approaches, filter-based methods represent some of the earliest attempts at battery RUL prediction. Zhang et al.[18] integrated the adaptive unscented Kalman filter into a three-dimensional open-circuit voltage model and reported an error of less than 2%. In ref[19], the Kalman filter was coupled with the Wiener process to account for parameter uncertainty, attaining enhanced performance relative to conventional methods. However, the dependence of these methods on accurate noise modeling and their sensitivity to initialization restricts their applicability under variable operating conditions. To address the shortcomings of filter-based methods, traditional machine learning algorithms have gained attention as a more flexible alternative. Patil et al.[20] developed classification and regression models for real-time RUL prediction by applying support vector machines (SVM) and deriving critical features. Qin et al.[21] proposed a hybrid Particle Swarm Optimization (PSO)- Support Vector Regression (SVR) model for RUL estimation across various failure thresholds; however, the scalability of this approach posed challenges when processing large volumes of data. An improved strategy combining the LMRLS algorithm with a radial basis correction-differential support vector machine (RBF-DSVM) has achieved accurate state of charge estimation for urban electric-vehicle batteries, with maximum errors of 0.037% and 0.336% under the HPPC and DST profiles[22]. In recent studies, tree-based algorithms including random forest and Extreme Gradient Boosting (XGBoost) have also been applied for RUL prediction[23, 24]. These methods achieved acceptable accuracy through hyperparameter optimization. However, their reliance on manual feature extraction and inability to capture complex temporal patterns paved the way for the emergence of deep learning approaches. The emergence of deep neural networks brought about a significant transformation in RUL prediction. Zhao et al.[25] utilized convolutional neural networks (CNN) for automatic feature extraction, whereas Ma et al.[26] mitigated the vanishing gradient problem using long short-term memory (LSTM) networks. Ardeshiri et al.[27] Also, combined gated recurrent units (GRU) with adaptive gradient descent and achieved satisfactory performance under nonlinear degradation conditions. Hybrid architecture has demonstrated superior performance. Fang et al.[28] enabled end-to-end prediction using solely current-voltage data by integrating CNN and LSTM. Tang and Yuan[29] enhanced accuracy under diverse charging conditions by integrating empirical mode decomposition with Bidirectional Gated Recurrent Unit (BiGRU). Similarly, Xia et al.[30] coupled complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) with gated recurrent unit and autoregressive integrated moving average (ARIMA) predictors, adding a residual-correction step. Their fusion model achieved RMSEs within 0.9% on NASA and 1.3% on CALCE, staying within 1.2% under cross-battery validation. Park et al.[31] introduced a multi-channel LSTM architecture that reduced the Mean Absolute Percentage Error (MAPE) by up to 63.7%. An online two-dimensional filtering framework [32] fused an Adaptive Taylor Kalman filter with an Improved Gaussian filter to suppress measurement noise online and extract more reliable aging features; unlike prior offline denoising

methods, it preserves real-time causality and is robust to battery inconsistency and temperature uncertainty, making it complementary to our end-to-end predictive framework. A robust multi-time-scale model combining singular filtering (SF), Gaussian process regression (GPR), and LSTM has been presented for whole-life-cycle remaining-capacity estimation, improving the low-temperature adaptability of lithium-ion batteries and achieving an RMSE of 2.3484%, MAE of 0.82526%, MAPE of 0.90716%, and R^2 of 92.457% while using only 55% of the data[33]. The development of attention mechanisms and Transformer networks marked a notable advancement in model enhancement. The Attention-Bidirectional Long Short-Term Memory (BiLSTM) model reduced the uncertainty associated with multi-step prediction[34], and the combination of Temporal Convolutional Network (TCN) with temporal attention decreased Root Mean Square Error (RMSE) on the NASA dataset by up to 33%[35]. The denoising Transformer neural network (DTNN) also improved accuracy and reliability by capturing long-range correlations[36]. More recent feature-centric studies have pushed this direction further: Xia et al. [37]reconstructed voltage with a second-order RC model to derive noise-free incremental-capacity (IC) and differential-voltage (DV) features, while Xia et al.[38] fused electrochemical and thermal features from constant-current (CC), differential thermal voltammetry (DTV), and differential thermal capacity (DTC) curves; both processed the resulting features with a single BiGRU network trained once on the complete life-cycle data. Despite these advances, deep learning models remain static after offline training and cannot adapt to real-time variations in operational conditions; a limitation that has directed researchers' attention toward digital twins. In response to these limitations, digital twin technology has emerged as a promising solution[39]. By creating a virtual counterpart of the battery system and continuously updating it with real-time data, this technology enables accurate simulation of battery health status and provides reliable RUL predictions. Unlike conventional digital models that are solely used for offline simulation and analysis, digital twins harness real-time data to facilitate real-time monitoring, predictive maintenance, and dynamic performance optimization[40]. These characteristics have positioned digital twins as an essential tool for connecting physical systems and cyberspace[41]. Furthermore, recent advancements in machine learning and the intelligent Internet of Things have enabled the seamless incorporation of digital twins into battery systems, opening new avenues for enhancing the prediction accuracy of lithium-ion battery parameters[42]. A comprehensive review of improved neural network algorithms (FNN, RNN, LSTM, GRU, CNN, and hybrids) for battery state-of-energy estimation in smart grids has been presented[43], examining the challenges of data sparsity, interpretability, and computational efficiency, and outlining future directions toward hybrid modeling and edge computing. In this regard, several studies have applied machine learning algorithms within a digital twin framework for battery state monitoring[44-46]. For instance, Jafari et al.[47] proposed a combination of XGBoost and an extended Kalman filter within a digital twin framework for battery capacity prediction. Deng et al.[48] integrated an equivalent circuit model and a convolutional neural network into a digital twin structure, generating synthetic data corresponding to various battery aging stages. Furthermore, Schmitt et al.[49] developed an encoder-decoder architecture for State of Health (SOH) estimation solely from BMS signals, without requiring capacity testing. More advanced data-driven frameworks have also been introduced for real-time battery capacity estimation, including the Adaptive Boosting (AdaBoost)-LSTM combination[50], a two-stage CNN-LSTM-Attention structure[51], and an LSTM equipped with a temporal attention mechanism achieving errors below one percent[52]. Despite these efforts, most existing frameworks have been trained on complete life cycle datasets and lack mechanisms for incremental updating with real-time data. Moreover, no comprehensive comparison among different deep learning architectures under identical conditions has been carried out,

and the capability of ensemble strategies for strengthening the robustness of long-term RUL prediction has not been sufficiently explored. Table 1 compares recent data-driven studies across five key criteria, showing that none jointly cover all of them, a gap this work addresses.

In this paper, to overcome the aforementioned limitations, a deep learning-based digital twin framework is proposed for predicting the remaining useful life of lithium-ion batteries. The proposed framework integrates five deep learning architectures, namely Stacked LSTM, TCN-LSTM, 1D-CNN-BiLSTM, Transformer-Encoder, and Conformer, within a digital twin environment. Unlike conventional approaches that train models offline on fixed datasets, the present framework adopts a cumulative incremental learning strategy, in which real-time data acquired from the physical battery are divided into sequential segments and progressively appended to the training set at each phase, enabling the model to adapt to new operational conditions continuously. In addition, six diverse ensemble strategies are employed to combine the outputs of the five models, thereby enhancing robustness and overall prediction accuracy. A thorough evaluation of the framework is performed on the NASA Prognostics Center of Excellence (PCoE) battery dataset[53] through a three-level evaluation protocol, and the results demonstrate a progressive improvement in prediction accuracy at each evaluation level. The generalization capability of the framework is further validated on the Oxford battery dataset under different operating conditions. To address the identified research gaps, the main contributions of this paper are summarized as follows:

- Proposing a digital twin framework in which five deep learning architectures Stacked LSTM, TCN-LSTM, 1D-CNN-BiLSTM, Transformer-Encoder, and Conformer are implemented as virtual twin models and systematically compared for real-time battery RUL prediction.
- Developing a cumulative incremental learning strategy that updates the digital twin in real time upon receiving new operational data, without losing the knowledge acquired from previous data.
- Conducting a sensitivity analysis to identify the optimal volume of real-time data at every model update stage; a total of 100 scenarios are executed and compared across the five architectures.
- Presenting an ensemble combination framework comprising six strategies for the systematic integration of the outputs from the five incremental deep learning models within the digital twin, aimed at enhancing the accuracy and robustness of RUL prediction.
- Designing a three-level evaluation protocol primarily conducted on the NASA dataset and further validated on the Oxford dataset, which quantitatively assesses the independent contribution of each proposed component: (a) the baseline model trained on historical data, (b) the enhanced model with real-time incremental updating, and (c) the integrated digital twin framework incorporating both incremental learning and ensemble strategies simultaneously.

The remainder of this paper is organized as follows: Section 2 reviews the theoretical foundations of battery digital twins. Section 3 describes the proposed framework, including the digital twin architecture, the real-time updating strategy, and the ensemble strategies. Section 4 presents the experimental results of the five deep learning architectures, along with sensitivity analysis

TABLE1. A comparative overview of the proposed framework against recent data-driven studies on battery health prediction

Ref	DT framework	multi-dataset	Real-time updating	Ensemble strategy	Multi-level evaluation
[37],[38]	✗	✓	✗	✗	✓
[30]	✗	✓	✗	✓	✗
[51],[52]	✓	✗	✓	✗	✗
[42],[44], [50]	✓	✗	✗	✗	✗
[46]	✓	✗	✗	✗	✗
[47]	✓	✗	✓	✗	✗
Proposed framework	✓	✓	✓	✓	✓

and the evaluation of the ensemble strategies' impact. Section 5 presents the conclusions and outlines future research directions.

2. Battery Digital Twin

A digital twin is a virtual representation of a physical system that models and forecasts the behavior of the system by receiving real-world data. This technology facilitates analysis, simulation, and performance optimization by constructing a digital model of a physical entity and continuously updating it with real-time data. Figure 1 illustrates the overall architecture of a digital twin for a battery. In this structure, historical data comprising voltage, current, and temperature are initially extracted from the physical twin and conveyed to the digital twin to construct the initial model. Then, real-time data is continuously streamed from the physical twin to the digital twin so that the digital model is updated in real time to capture the actual state of the battery. Finally, the digital twin returns its predictions and insights to the physical system as control feedback. The physical layer of this architecture comprises the lithium-ion battery and its management system, which measures and transmits operational data to the digital twin.

2.1. Battery Management System

The battery management system serves a critical function in maintaining the safe and stable operation of lithium-ion batteries. This system prevents critical conditions such as overcharging and overdischarging by monitoring vital parameters in real time and applying control strategies. A standard BMS is composed of several subsystems, each tasked with a specific function. As shown in Figure 2, these subsystems include: battery pack, state measurement, state estimation, state management, fault diagnosis, and communication module. Moreover, the choice of BMS topology (centralized, modular, or decentralized) directly affects state estimation accuracy[54].

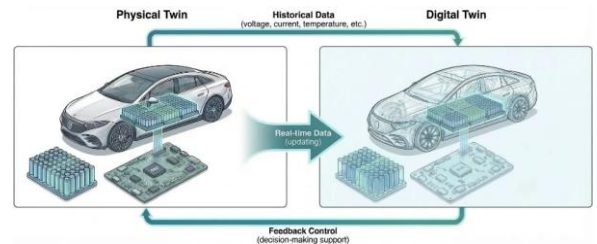


Figure 1: Battery Digital Twin Architecture Based on Data Exchange

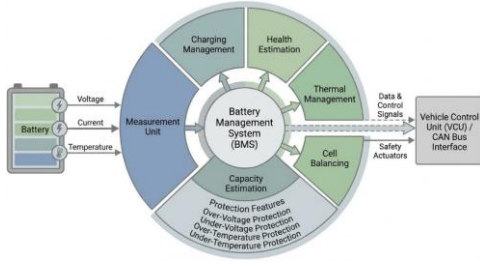


Figure 2: Battery Management System Architecture

In the battery pack structure, cells are organized in a series-parallel arrangement into multiple modules, enabling the management of several thousand cells in a single system. The state measurement unit is tasked with collecting the voltage, current, and temperature of the cells from sensors and relaying this data to other system components. These data are utilized by the state estimation unit to compute key RUL indicators, which require advanced algorithms because they cannot be measured directly.

State management consists of three components: cell balancing to correct charge differences between cells in a battery pack, thermal management to maintain operating temperatures within 20 to 40 degrees Celsius by activating cooling or heating systems, and charge/discharge management to control the phase difference between voltage and current to improve power factor and increase energy efficiency. The fault diagnosis unit continuously monitors behavioral patterns to identify anomalies caused by structural defects or environmental conditions before they propagate. Finally, the communication module transmits the processed data, including total voltage, current, and cell temperature-voltage limit values, to the vehicle control unit, ensuring standardized and reliable communication. This integrated structure, with subsystem coordination, ensures optimal and stable battery performance throughout the life cycle.

2.2. Battery State Estimation

Battery state estimation, as one of the core functions of the BMS, is tasked with quantitatively evaluating the current condition and forecasting the future behavior of the cell. Among the various aspects of this estimation, the RUL is a prognostic indicator that quantifies the count of remaining operable cycles before the End-of-Life (EOL) point. Accurate RUL prediction enables optimal scheduling of cell replacement and effective lifecycle management of the battery pack. From a quantitative perspective, cycle-based RUL is defined as follows:

$$RUL(t) = t_{EOL} - t \quad (1)$$

where t_{EOL} denotes the number of cycles at the end-of-life point and represents the present cycle of the battery. In this study, the EOL criterion is defined at 70% of the initial nominal capacity, i.e., 1.4 Ah. Since capacity fade directly mirrors the cumulative electrochemical degradation within the cell, this threshold provides a dependable criterion for identifying the operational end-of-life point of the battery.

3. Methodology

3.1. Dataset Description and Preprocessing

To validate the proposed method, the NASA Prognostics Center of Excellence (PCoE) battery dataset was used. This dataset contains experimental measurements of 18650 lithium-ion batteries with a capacity of 2.0 Ah subjected to charge/discharge cycles. The batteries were charged using the constant current-constant voltage (CC-CV) method and discharged with constant current at a constant temperature

of 24°C. Test cycles continued until reaching end-of-life (EOL), equivalent to a 30% capacity reduction. The parameters of voltage, current, temperature, and discharge capacity were recorded in each cycle, which provided the basis for developing RUL prediction models. To further assess generalizability, an additional dataset was employed: the Oxford Battery Degradation Dataset (Battery Intelligence Lab, University of Oxford), which consists of Kokam pouch cells with a rated capacity of 740 mAh, aged within a thermal chamber at a constant temperature of 40°C. The cells were discharged under a dynamic profile derived from the Urban Artemis driving cycle and charged via a CC-CV protocol, with characterization performed every 100 cycles to extract capacity. This dataset complements the constant-temperature NASA data under elevated-temperature and dynamic-load conditions.

The raw data required careful preprocessing to achieve appropriate quality and consistency for training deep learning models. First, a data cleaning process was performed to identify and process any irrelevant or missing data and ensure the consistency of the dataset. This step was necessary to preserve the integrity and dependability of the laboratory data. After cleaning, Z-score normalization was performed for all input parameters to mitigate the influence of differences in varying scales on the model learning process. This method maps the data into a normalized space characterized by a mean of zero and a variance of one, employing the mean (μ) and standard deviation (σ) of each feature:

$$z = \frac{x - \mu}{\sigma} \quad (2)$$

Where x is the original data value, μ is the mean, and σ is the standard deviation of that feature. This process ensures that all input variables fall within the same range and the model is equally sensitive to all features, which leads to improved numerical stability and convergence speed of the neural network.

3.2. Proposed Digital Twin Framework

The digital twin framework proposed in this study is designed to construct an intelligent virtual replica of a real lithium-ion battery from the NASA PCoE dataset. Unlike static modeling approaches that rely solely on historical data, this framework integrates three types of data, namely historical training data, offline test data, and real-time operational data, within a unified and continuous process, so that the prediction accuracy of the model progressively improves over time. This framework consists of three main components: (1) the overall model architecture, which employs a variety of deep learning models, spanning recurrent, convolutional, attention-based, and hybrid architectures, to map multidimensional battery measurements to capacity estimation; (2) a real-time model updating strategy grounded in Cumulative Incremental Learning, which allows the digital twin to adjust to new operational conditions without discarding previously learned patterns; and (3) an Ensemble framework comprising six integration strategies, which methodically combines the outputs of five incremental models and exploits architectural diversity to deliver a more stable and accurate final RUL prediction. In addition, the Oxford dataset is employed for further validation, assessing the generalization capability of the proposed framework under different temperature operating conditions. The subsequent subsections present a detailed description of each of these components.

3.2.1. Overall Architecture

The foundation of the proposed digital twin modeling is built upon a comparative deep learning approach, in which five distinct time series processing architectures are trained, validated, and evaluated under identical experimental conditions. Maintaining uniform preprocessing stages, training conditions, and evaluation criteria across all five

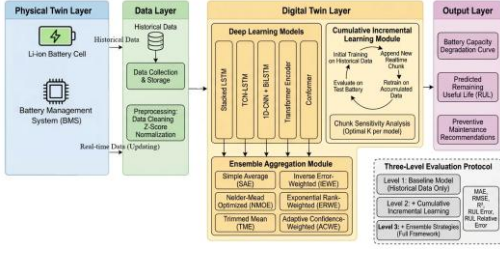


Figure 3: Proposed Intelligent Digital Twin Framework for Lithium-Ion Battery Condition Evaluation and RUL Prediction

models guarantees that any differences in prediction accuracy solely represent the inherent differences among the architectures themselves. The overall architecture of the proposed framework consists of four main layers that operate in interaction with one another:

Physical Twin Layer: This layer encompasses the actual components of the battery system, including lithium-ion cells, the Battery Management System (BMS), and associated sensors. Five operational parameters, namely voltage, current, temperature, time, and charge/discharge status indicator, are continuously extracted from this layer.

Data Layer: This layer serves as an interface between the physical twin and the digital twin. Its primary function is to collect and store the raw operational data received from the physical layer and to provide this data in a structured format for processing in the digital twin layer.

Digital Twin Layer: This layer represents the digital reflection of the physical battery system components, designed to simulate the actual behavior of the system and to fulfill the prediction objectives. Within this layer, five distinct data-driven deep learning architectures, including Stacked LSTM, TCN-LSTM, 1D-CNN-BiLSTM, Transformer-Encoder, and Conformer, are employed. These architectures cover a broad spectrum of time series modeling approaches, ranging from recurrent networks and convolutional combinations to attention mechanisms and transformer-based models. This architectural diversity enables a comprehensive evaluation of each modeling approach in extracting battery degradation patterns and predicting the RUL.

Output Layer: This layer presents the final results, including the battery capacity degradation process, predicted value of the RUL, and preventive maintenance recommendations, thus supporting informed decision-making at the operational level.

The architecture of the proposed battery digital twin is depicted in Figure 3. Since the accuracy of RUL prediction is closely tied to the selection and design of the predictive model architectures, the internal structure of each of the five architectures is elaborated in the following.

The first architecture implemented in the digital twin layer is the Stacked LSTM network. This architecture was selected owing to the inherent capability of LSTM networks in capturing long-term temporal dependencies, which is vital for recognizing the gradual capacity degradation patterns in lithium-ion batteries. Moreover, the gating mechanism of LSTM cells effectively addresses the vanishing gradient problem and supports stable learning from long charge-discharge sequences. The operation of each LSTM cell at time step t

is defined through a set of control gates that regulate the flow of information within the cell memory. The computational equations of these gates are expressed as follows[55]:

$$f_t = \sigma(w_f[h_{t-1}, x_t] + b_f) \quad (3)$$

$$i_t = \sigma(w_i[h_{t-1}, x_t] + b_i) \quad (4)$$

$$g_t = \delta(w_g[h_{t-1}, x_t] + b_g) \quad (5)$$

$$o_t = \sigma(w_o[h_{t-1}, x_t] + b_o) \quad (6)$$

$$S_t = f_t \cdot S_{t-1} + i_t \cdot g_t \quad (7)$$

$$h_t = o_t \cdot \delta(S_t) \quad (8)$$

where f_t denotes the forget gate, i_t the input gate, g_t the candidate memory vector, o_t the output gate, S_t the memory cell state, and h_t the hidden state output vector at time step t . Additionally x_t represents the input vector at time step t , w the learnable weight matrix, and b the bias vector corresponding to each gate. The activation function σ denotes the sigmoid function and δ denotes the hyperbolic tangent function, which are defined as follows:

$$\sigma(x) = \text{sigmoid}(x) = \frac{1}{1 + \exp(-x)} \quad (9)$$

$$\delta(x) = \tanh(x) = \frac{\exp(x) - \exp(-x)}{\exp(x) + \exp(-x)} \quad (10)$$

This model employs a stacked configuration consisting of two consecutive LSTM layers, both utilizing the tanh activation function. The first LSTM layer propagates its hidden state vectors across all time steps, allowing the second layer to operate on a more complete temporal representation. The second LSTM layer compresses the entire input sequence into a fixed-length vector, which serves as a compact representation of the battery degradation trajectory. After each LSTM layer, a Dropout layer is applied, which randomly deactivates a portion of neurons during training, thereby enhancing the generalization capability of the model and preventing overfitting to the training data. The extracted temporal representation is then fed into a fully connected Dense layer with ReLU activation, which performs a nonlinear mapping to an intermediate feature space. This design choice enables the network to capture complex and non-trivial relationships between the extracted temporal features and the target RUL value. Following a final Dropout layer, the output layer composed of a single neuron with linear activation produces the predicted RUL as a continuous numerical value. The optimization process is performed using the Adam algorithm, and gradient clipping is applied to stabilize the training process and prevent gradient explosion.

The second architecture is the hybrid TCN-LSTM network, designed to simultaneously leverage the capabilities of Temporal Convolutional Networks and LSTM networks. The core idea behind this combination is that the temporal convolutional layers identify local and short-term patterns within the sensor sequences. In contrast, the LSTM layer subsequently models long-term dependencies over the extracted representations.

The temporal convolutional component of this model consists of two Conv1D layers with causal padding and ReLU activation. The use of causal padding ensures that the output at each time step depends solely on the information from the current and past time steps and prevents any information leakage. The dilation rate in these two layers is set incrementally; the first layer, with a dilation rate of one, extracts adjacent and local patterns, while the second layer, with a dilation rate of two, expands the receptive field of the network without increasing the number of parameters, enabling the detection of temporal patterns

at a larger scale. The causal dilated convolution operation at time step s is defined as follows[56]:

$$F(s) = \sum_{i=0}^{k-1} f(i) \cdot x_{s-di} \quad (11)$$

where f denotes the convolutional filter with size k , d the dilation rate, and x_{s-di} the input value at temporal position $s - di$. The effective receptive field of the network for L convolutional layers is obtained from the following relation:

$$RF = 1 + \sum_{l=1}^L (k_l - 1) \cdot d_l \quad (12)$$

After each convolutional layer, a Dropout layer is applied to prevent overfitting. The output of the convolutional section, a sequence of temporal feature maps, is fed into an LSTM layer with tanh activation. The internal operation of this LSTM layer follows equations (3) through (8) presented in the previous model. This layer compresses the entire sequence of convolutional representations into a fixed-length vector, retaining long-term temporal dependencies among the extracted features within its internal memory. Following the Dropout application, the resulting vector is passed to a fully connected layer with ReLU activation, which performs a nonlinear mapping to an intermediate feature space. The ReLU activation function is defined as follows:

$$ReLU(z) = \max(0, z) \quad (13)$$

Finally, the output layer, composed of a single neuron with linear activation, generates the predicted RUL value. The model is optimized using the Adam algorithm with gradient clipping employed.

The third architecture is a hybrid 1D-CNN-BiLSTM network equipped with an Attention mechanism. This model is designed to simultaneously exploit hierarchical extraction of spatiotemporal features and bidirectional modeling of temporal dependencies, representing a step beyond the two preceding architectures in terms of depth and structural complexity. Unlike the TCN-LSTM model, which employs a simple sequential structure, this model is implemented using the Functional API, enabling the definition of more complex data flows and the integration of the Attention mechanism. The convolutional component of this model consists of three successive Conv1D layers with ReLU activation and same padding. After each convolutional layer, Batch Normalization is applied, which accelerates the convergence process and enhances training stability by normalizing the distribution of activations within each training batch. In the second convolutional layer, the number of filters is increased, followed by a Max Pooling layer that reduces the temporal dimensions while preserving the features of the most prominent local patterns. The third convolutional layer further refines the representations and prepares the final feature maps for input to the recurrent component. Dropout layers are applied throughout this section to regularize the model.

The output of the convolutional component is passed into two Bidirectional LSTM layers, both employing tanh activation. The bidirectional structure allows the model to process the input sequence simultaneously in both forward and backward directions, offering a richer context from both past and future at each time step. The output of the bidirectional layer at each time step t is derived by concatenating the hidden state vectors from the forward and backward passes[57]:

$$h_t^f = H(h_{t-1}^f, x_t) \quad (14)$$

$$h_t^b = H(h_{t+1}^b, x_t) \quad (15)$$

$$h_t = [h_t^f; h_t^b] \quad (16)$$

where H denotes the LSTM unit mapping function, h_t^f the hidden state of the forward pass, h_t^b the hidden state of the backward pass, and the

concatenation operator merges the vectors from both directions. BiLSTM layers propagate their hidden state vectors across all time steps so that the complete sequence of representations remains available for the Attention mechanism.

A Self-Attention mechanism is applied to the output of the last BiLSTM layer, allowing the model to adaptively assign greater weight to time steps that carry more critical information regarding capacity degradation. The scaled attention computation is defined as follows[58]:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (17)$$

where Q , K , and V represent the Query, Key, and Value matrices, respectively, and d_k denotes the dimensionality of the key vectors. In this model, since the Attention mechanism is employed in a Self-Attention manner, all three matrices are obtained from the output of the BiLSTM layer. Subsequently, a Global Average Pooling layer transforms the resulting weighted sequence into a fixed-length vector. This vector is then fed through three successive fully connected layers with ReLU activation and Dropout, performing a nonlinear mapping of the feature vector to the output regression space across multiple stages. The output layer, composed of a single neuron with linear activation, generates the predicted RUL value. The model is optimized using the Adam algorithm.

The fourth architecture is the Transformer-Encoder, which entirely abandons recurrent and convolutional structures and relies solely on the Attention mechanism as the core component for modeling temporal dependencies. This approach enables parallel processing of all time steps. It provides each step with direct access to the information of all other steps, without the inherent limitations of short-term memory.

In the first step, the three-dimensional input is mapped to the model's representation space through a fully connected layer without an activation function. This linear projection transforms the input feature dimensions to the dimensionality required by the Transformer block without introducing any nonlinearity. A Sinusoidal Positional Encoding is then added to this representation. Since the Attention mechanism alone is incapable of recognizing the position of each step within the input sequence, this encoding injects temporal order information into the representation based on sine and cosine functions with varying frequencies [58]:

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) \quad (18)$$

$$PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right) \quad (19)$$

where pos denotes the position of the time step within the sequence, i the dimension index, and d_{model} the dimensionality of the model's representation space. The core of the model is composed of a single Transformer block that adheres to the general architecture of the original encoder. In the first part of the block, a Single-Head Self-Attention mechanism is applied in a Self-Attention manner. The output of this mechanism, after passing through Dropout, is combined with the original block input via a Residual Connection, followed by Layer Normalization. Residual connections promote gradient flow during training and alleviate the vanishing gradient problem. The output of this stage is formulated as follows, where the residual connection adds the input to the sub-layer output before normalization:

$$z = \text{LayerNorm}(x + \text{Dropout}(\text{Attention}(x))), \text{LayerNorm}(u) \\ = \gamma \odot \frac{u - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta \quad (20)$$

where μ and σ^2 denote the mean and variance computed over the feature dimension, γ and β are learnable scale and shift parameters, and ϵ (here 10^{-6}) ensures numerical stability. In the second part of the block, a Feed-Forward Network comprising two fully connected layers is utilized; the first layer with ReLU activation projects the representation into a higher-dimensional space, whereas the second layer without an activation function maps it back to the original dimensions. This two-layer transformation is expressed as:

$$\text{FFN}(z) = \max(0, zW_1 + b_1)W_2 + b_2 \quad (21)$$

where $W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ and $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$ are the expansion and projection weight matrices, b_1, b_2 the corresponding biases, and $\max(0, x)$ denotes the ReLU activation. The output of this network is likewise regulated through a residual connection and a second Layer Normalization. The output of the Transformer block is transformed into a fixed-length vector through a Global Average Pooling layer and, after passing through a fully connected layer with ReLU activation and Dropout, is directed into the output layer with linear activation to generate the predicted RUL value. The model is optimized using the Adam algorithm.

The fifth and most complex architecture is the Conformer, which attempts to complementarily leverage the strengths of both approaches by simultaneously combining the Attention mechanism and convolution within a unified structure. This architecture is capable of concurrently capturing both global long-range dependencies through Attention and local patterns through convolution.

In the first step, an FFT Feature Extractor module is applied to the input. This module extracts the frequency-domain representation of the signal using the Fast Fourier Transform, and concatenates the resulting real and imaginary components with the original input. The combined vector is then mapped to the model's representation space through a fully connected layer without an activation function. This operation is formulated as follows:

$$X_f = \text{FFT}(x), \quad z = [x \| \text{Re}(X_f) \| \text{Im}(X_f)]W_e + b_e \quad (22)$$

where $\text{FFT}(\cdot)$ denotes the Fast Fourier Transform, $\text{Re}(\cdot)$ and $\text{Im}(\cdot)$ are the real and imaginary parts, and W_e, b_e are the weight matrix and bias of the fully connected embedding layer. Subsequently, Sinusoidal Positional Encoding is added to the output to provide the model with temporal order information. The core of the model consists of two consecutive Conformer blocks. Each block comprises four sub-modules that are executed in the following order: first, a Feed-Forward Network with a Half-step Residual connection is applied, which contains Layer Normalization, a fully connected layer with Swish activation, Dropout, a fully connected layer without activation, and a second Dropout. The Swish activation function is formulated as follows[59]:

$$\text{Swish}(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}} \quad (23)$$

where $\sigma(x)$ is the Sigmoid function. Unlike ReLU, this function is smooth and non-monotonic and has demonstrated superior performance in deep networks. The output of the Feed-Forward module is incorporated into the input through a Half-step Residual connection[60]:

$$y = x + \frac{1}{2} \cdot \text{FFN}(x) \quad (24)$$

where x is the sub-module input and $\text{FFN}(x)$ is the output of the Feed-Forward Network. The coefficient $1/2$ stabilizes training in structures that contain two FFN modules per block.

Next, the Multi-Head Self-Attention mechanism is employed, where the input, after Layer Normalization, enters the Attention module, and its output, following Dropout, is added to the input through a full Residual connection. The third sub-module is the Conformer Convolution module, in which, following Layer Normalization, a Pointwise Convolution doubles the number of channels, and then a Gated Linear Unit (GLU) mechanism is applied, where half of the channels serve as the input and the other half, after passing through the Sigmoid function, act as the gate, with their element-wise product forming the output. This gating mechanism is expressed as:

$$\text{GLU}(a, b) = a \odot \sigma(b) \quad (25)$$

where a and b are the two halves of the pointwise-convolution output, $\sigma(\cdot)$ is the Sigmoid function, and $\odot$ denotes element-wise multiplication. Subsequently, a standard convolution followed by Swish activation is applied, and another Pointwise Convolution restores the dimensions to their original size; this module is also accompanied by a full Residual connection. Finally, another Feed-Forward Network with the same Half-step structure and a final Layer Normalization is applied to the block output. After passing through the two Conformer blocks, the output is compressed into a fixed-size vector by a GRU layer, and after passing through a fully connected layer with Swish activation and Dropout, the output layer with linear activation produces the predicted RUL value. The model is optimized using the Adam algorithm.

The training process of all architectures described in this layer follows a cumulative incremental learning strategy. In the first phase, each model is initially trained on historical data, and in subsequent phases, as real-time data is progressively obtained from the physical twin, the new data is appended to the existing training set, and the model is updated on the entire accumulated dataset. This approach preserves the model's prior knowledge acquired from historical data while enabling its adaptation to new operational conditions.

3.2.2. Real-Time Model Update Strategy

One of the fundamental characteristics of the digital twin framework, which distinguishes it from conventional predictive models, is the capability of continuous synchronization of the virtual model with the physical asset through operational data streams. As illustrated in Figure 1, the virtual side of the battery digital twin must continuously receive operational data from the physical side and be updated accordingly, so as to maintain an accurate and up-to-date representation of the system's evolving state. In the proposed framework, the data-driven models of the digital twin layer are updated in real time upon receiving new operational data from the physical twin.

To this end, the training and updating process of the models follows a Cumulative Incremental Learning strategy. In the first phase (Phase Zero), each model is trained solely on the historical data of a reference battery. Subsequently, as real-time data is gradually received from the physical twin, these data are divided into sequential chunks, and in each phase, a new chunk is appended to the existing training set. After the addition of each chunk, the model is retrained on the entire accumulated data, which includes the initial historical data and all chunks received up to that point, and is evaluated on an independent test battery. This design ensures that the model's prior knowledge is preserved while its adaptability to new operational conditions is progressively enhanced.

The key parameter governing this strategy is the number of chunks K into which the real-time data are divided. Smaller values of K result in larger chunks and fewer but more voluminous updates, whereas larger values of K lead to more frequent but finer-grained updates. To determine the optimal value of this parameter, a comprehensive sensitivity analysis has been designed in which each of the five architectures is independently evaluated with $K = 1, 2, \dots, 20$. Based

on the results of this analysis, the optimal value of K is determined for each architecture and employed in the real-time updating process of the digital twin.

3.2.3. Ensemble Aggregation Strategies

Combining the outputs of multiple deep learning models into a single prediction is an effective approach for improving overall prediction accuracy by leveraging the strengths and complementary characteristics of different models. After each of the five base deep learning models is trained under the digital twin framework and updated through the cumulative incremental learning strategy, the prediction output of each model becomes independently available. Although each model is individually capable of estimating the RUL, the inherent diversity of the architecture causes each model to exhibit distinct patterns of error and accuracy in its predictions. The ensemble layer in the proposed framework is designed to intelligently integrate these diverse outputs and achieve a more accurate estimate. To this end, six ensemble strategies have been employed: Simple Average Ensemble (SAE), Inverse Error-Weighted Ensemble (IEWE), Nelder-Mead Optimized Ensemble (NMOE), Exponential Rank-Weighted Ensemble (ERWE), Trimmed Mean Ensemble (TME), and Adaptive Confidence-Weighted Ensemble (ACWE). The general form of the weighted combination for all strategies is defined as follows:

$$\hat{y}_{\text{ens}}(i) = \sum_{m=1}^M \omega_m(i) \hat{y}_m(i), \quad \sum_{m=1}^M \omega_m(i) = 1, \quad \omega_m(i) \geq 0 \quad (26)$$

where $\hat{y}_{\text{ens}}(i)$ is the final ensemble prediction for sample i , $\hat{y}_m(i)$ is the prediction of the m -th model for the same sample, and ω_m is the weight assigned to the m -th model, which, depending on the strategy, is either constant or sample-dependent on i . What distinguishes the six strategies from one another is the manner in which the weights ω_m are computed.

SAE assigns a uniform weight of $\omega_m = 1/M$ to all models without incorporating any performance-related information, thereby providing the most basic form of prediction combination and allowing all models to contribute equally to the final output regardless of their individual accuracy.

IEWE determines the weight of each model in proportion to the inverse of its error on the validation data:

$$\omega_m = \frac{(e_m + \epsilon)^{-1}}{\sum_{j=1}^M (e_j + \epsilon)^{-1}} \quad (27)$$

where e_m is the prediction error of the m -th model on the validation data, and ϵ is a numerical stabilization coefficient. These weights are recalculated after each incremental learning step; consequently, models with lower errors receive a greater share in the final prediction.

NMOE determines the optimal weight vector directly by minimizing the combination error. To this end, the Nelder-Mead simplex method is executed with 50 random starting points sampled from a Dirichlet distribution to identify the weights that minimize the ensemble prediction error on the validation data. Unlike IEWE, which assumes an inverse relationship between error and weight, this approach extracts the best combination directly from the data.

ERWE determines the weights based on the error ranking of the models using an exponentially decaying function:

$$\omega_m = \frac{\exp(-0.5 r_m)}{\sum_{j=1}^M \exp(-0.5 r_j)} \quad (28)$$

where $r_m \in \{0, \dots, M-1\}$ is the rank of the m -th model. Unlike IEWE, this strategy is not sensitive to the absolute error values and only

considers the relative ordering of the models. As a result, when the error scales differ substantially across models, it prevents any single model from excessively dominating the final combination.

TME removes the minimum and maximum predictions for each sample and averages the remaining $M-2$ values. This eliminates the effect of outlier predictions, which some models may produce during incremental learning, without requiring any weight computation.

ACWE implements adaptive weighting at the individual sample level:

$$\omega_m(i) = \frac{\alpha_m(i) \cdot \beta_m}{\sum_{j=1}^M \alpha_j(i) \cdot \beta_j} \quad (29)$$

where $\alpha_m(i) = (|\hat{y}_m(i) - \text{median}(\hat{y}(i))| + \epsilon)^{-1}$ represents the local confidence of the model at sample i , and $\beta_m = (e_m + \epsilon)^{-1}$ represents its global reliability. The rationale behind ACWE is that confidence in a model depends not only on its overall performance but also on its behavior at each sample; a model that produces a prediction significantly different from the consensus of the other models at a given sample should receive a lower weight for that sample.

These six strategies span from simple averaging to sample-level adaptive weighting, enabling a comprehensive comparison of combination approaches in the results section.

3.3. Evaluation Metrics

To assess the accuracy and efficiency of the proposed approach in estimating key battery parameters, three standard evaluation metrics have been employed. The mean absolute error (MAE) calculates the difference between the estimated and actual values and directly indicates how much the model deviates from the experimental data on average:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |l(i) - \hat{l}(i)| \quad (30)$$

where $l(i)$ is the actual value and $\hat{l}(i)$ is the predicted value. The root mean square error (RMSE), by squaring the errors, assigns greater weight to large deviations and exhibits higher sensitivity to significant errors:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (l(i) - \hat{l}(i))^2} \quad (31)$$

R^2 evaluates the degree of model fit to the observed data and determines what proportion of the variability in the dependent variable can be explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^N (l(i) - \hat{l}(i))^2}{\sum_{i=1}^N (l(i) - \bar{l})^2} \quad (32)$$

MAE and RMSE values close to zero, along with R^2 close to one, indicate a closer fit of the model to the experimental data and a stronger fit to the actual observations. In addition to the three metrics described above, two more specialized indicators have been adopted to better capture the prediction performance in terms of remaining useful life. The RUL absolute error (RUL_{ae}) measures how far the predicted value deviates from the true value, regardless of the direction of the error:

$$RUL_{ae} = |RUL_{pre} - RUL_{rea}| \quad (33)$$

The RUL absolute relative error (RUL_{ape}) expresses this deviation as a percentage of the actual value, which allows for a fairer comparison across batteries with different life spans.

$$RUL_{ape} = \frac{|RUL_{pre} - RUL_{rea}|}{RUL_{rea}} \times 100\% \quad (34)$$

where RUL_{pre} and RUL_{rea} denote the predicted and actual remaining useful life, respectively. Lower values of RUL_{ae} and RUL_{ape} correspond to more accurate predictions.

The combined use of all five metrics offers a well-rounded evaluation of model performance. MAE and RMSE quantify the prediction error in absolute terms, R^2 reflects the overall goodness of fit, and RUL_{ae} together with RUL_{ape} specifically assess the accuracy of remaining useful life estimation in both absolute and relative terms.

4. Results and Discussion

In this section, the results of implementing and evaluating the proposed deep learning-based digital twin framework for capacity prediction and RUL prediction of lithium-ion batteries are comprehensively presented and analyzed. The evaluation is conducted at three complementary levels. At the first level, five deep learning architectures are trained solely on historical capacity degradation data, and the ability of each architecture to predict the capacity fade trend and forecast the RUL is assessed; the objective at this level is to identify the inherent strengths and weaknesses of each architecture before any new operational data is introduced. At the second level, the trained models are incrementally updated with real-time data to evaluate the adaptability of each architecture to new data and the extent of performance improvement achieved through incremental learning. At the third level, the outputs of the base models are combined through various ensemble approaches to investigate the potential of simultaneously leveraging the advantages of multiple diverse architectures. Across all three levels, the performance of each approach is evaluated in terms of both capacity prediction and RUL prediction accuracy.

4.1. Offline Model Evaluation and Predictive Performance

The offline evaluation results of the five deep learning architectures reveal notable differences in how each model predicts the capacity degradation trend of lithium-ion batteries. These differences are observed both in the accuracy of predicting the overall capacity degradation trend and in handling sudden, short-term capacity fluctuations. Figure 4 presents the capacity prediction results of each model compared to the actual measured capacity over successive charge-discharge cycles, and Figure 5 provides a quantitative comparison of the five models based on three evaluation metrics: MAE, RMSE, and R^2 .

Stacked LSTM, composed of two recurrent layers with 64 units each and a Dropout mechanism, captured the overall downward trend of capacity degradation with reasonable accuracy and produced stable predictions that were robust against measurement fluctuations. As shown in Figure 4, the predicted trajectory exhibits acceptable alignment with the actual values, particularly in regions characterized by a nearly linear capacity fade. The multi-layer recurrent structure of this architecture propagates temporal information sequentially and inherently smooths out measurement noise. While this characteristic ensures the stability of the predicted trajectory, it constrains the model's ability to respond to short-term variations and transient capacity behavior, especially in the mid-life region of the cycle life,

leading to slightly overestimated predictions relative to the actual values.

This limitation arises because the recurrent architecture assigns approximately equal weight to all preceding time steps, thereby favoring long-term prediction stability over tracking instantaneous capacity changes. TCN-LSTM, as the first hybrid architecture evaluated, combines one-dimensional convolutional layers with an LSTM layer to perform local temporal feature extraction and sequential dependency modeling hierarchically. As illustrated in Figure 4, this model reproduces the overall downward capacity trend with adequate accuracy; however, in intervals where the capacity behavior is accompanied by short-term fluctuations, the predicted trajectory does not fully follow the actual variations, and noticeable deviations are observed. This limitation indicates that the sequential aggregation of information within the recurrent layer, while ensuring prediction stability, constrains the model's ability to track sudden capacity changes. 1D-CNN-BiLSTM, by combining hierarchical feature extraction through three convolutional layers with 128, 256, and 128 filters, bidirectional temporal processing with BiLSTM units of 256 and 128, and an attention mechanism, can capture more complex temporal dependencies across multiple scales. The tapering structure of the Dense layers with 256, 128, and 64 units further enables a gradual compression of the learned representations. This model follows the overall degradation trend effectively; however, the attention mechanism, by focusing on dominant patterns, slightly reduces the model's sensitivity to minor capacity fluctuations. Transformer Encoder, with an MAE of 0.062, RMSE of 0.089, and R^2 of 0.874, achieved a notable performance among the five evaluated architectures. Despite having a considerably smaller number of parameters compared to the other architectures, this model reproduces the overall capacity degradation trend with adequate accuracy. The attention mechanism, by assigning differential weights to different time steps, in contrast to the approximately uniform weighting inherent in recurrent architectures, enables the model to process time

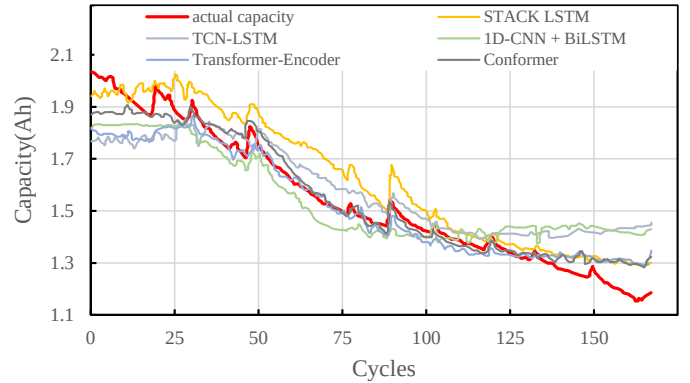


Figure 4: Offline Capacity Prediction Results of Deep Learning Models

TABLE2. Optimal Hyperparameters of the Five Architectures for Initial Training and Real-Time Updating Phases

Model	Architecture Specific Parameters	Dense Units	Dropout Rates	Learning Rate	Epochs (Train / Update)
Stacked LSTM	LSTM: 2 layers / 64 units	32	0.15	0.0015	600 / 50
TCN-LSTM	Conv1D: 32, Kernel: 7, Dilation: (1,2), LSTM: 64	16	0.3 / 0.4	0.001	400 / 50
1D-CNN-BiLSTM	Conv1D: 128,256,128, Kernel: 3, Pool: 2, BiLSTM: 256,128	256, 128, 64	0.2–0.3 / 0.3 / 0.2–0.4	0.001	400 / 50
Transformer Encoder	d_{model} : 16, h : 1, d_{ff} : 32	16	0.1 / 0.2 (Att / Dense)	0.001	200 / 50

Conformer	d_{model} : 24, h : 3, d_{ff} : 48, Conv kernel: 5, GRU: 12	16	0.1	0.0005	100 / 50
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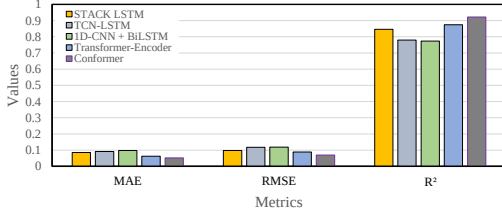


Figure 5: Performance Comparison of Offline Prediction Models

steps containing more critical information with greater emphasis. This characteristic proves particularly effective in simultaneously identifying long-term gradual capacity fade patterns and relatively faster variations. Nevertheless, the compact dimensionality of the representation space limits the model's representational capacity, which slightly diminishes its ability to track fine-grained capacity changes in intervals exhibiting complex and nonlinear behavior. Conformer, with an MAE of 0.052, RMSE of 0.07, and R^2 of 0.922, achieves the best performance among the five evaluated architectures. The predicted trajectory of this model exhibits very close alignment with the actual capacity values throughout the entire cycle of life and is capable of accurately tracking both the overall capacity fade trend and short-term fluctuations. This high accuracy stems from the complementary combination of frequency-domain pattern extraction by the FFT Feature Extractor, long-range dependency modeling by Multi-Head Self-Attention, and local pattern recognition by the convolutional module with GLU. The gating mechanism of GLU enables the model to regulate feature weights and separate relevant information from noise adaptively. The GRU layer further assists the model by compressing temporal information, allowing it to predict capacity changes uniformly across different regions of the cycle life. The quantitative comparison of the five architectures confirms the superiority of Conformer across all three evaluation metrics. It demonstrates that the integration of attention, convolutional, and recurrent mechanisms within a unified architecture significantly enhances the capacity to model complex capacity degradation patterns. Transformer Encoder, despite having fewer parameters, delivers acceptable performance, yet its accuracy is limited by the absence of local feature extraction and temporal compression mechanisms. Purely recurrent architectures such as Stacked LSTM, as well as hybrid architectures such as TCN-LSTM and 1D-CNN-BiLSTM, although they follow the overall degradation trend effectively, face limitations in tracking short-term changes and capacity fluctuations. The offline evaluation results on historical data indicate that all five architectures are capable of learning the principal capacity degradation patterns and provide the necessary foundation for the subsequent step.

4.2. Sensitivity Analysis on Real-time Data Chunks

One of the fundamental challenges in implementing a digital twin is maintaining continuous synchronization between the digital twin and its physical counterpart. Since lithium-ion batteries are subject to factors such as temperature fluctuations, variations in charge/discharge rates, and electrochemical aging phenomena throughout their operational lifetime, their capacity degradation patterns gradually evolve. Consequently, a model built solely on historical data will progressively lose its accuracy and diverge from reality. This necessitates a real-time updating mechanism to preserve the accuracy and reliability of the digital twin. To this end, a cumulative incremental learning approach has been adopted, in which at each update stage, all previously available data along with the newly acquired segment are incorporated into the training set. In this manner, the model retains its prior knowledge while simultaneously adapting to emerging degradation patterns. To implement this process, the real-

time data are divided into K consecutive segments, and a sensitivity analysis is performed for $K=1$ through $K=20$. The upper limit of K at twenty stems from practical observations, since beyond this value, the cumulative computational cost, including training time, memory usage, and model storage overhead, grows disproportionately with no noticeable gain in prediction accuracy. Accordingly, the goal of this analysis is to identify the optimal value of K that achieves the best trade-off between prediction accuracy and computational cost for each architecture.

The optimal hyperparameters for each of the five architectures under investigation are reported in Table 2. Figure 6 depicts the capacity prediction results of each architecture at its corresponding best K value under real-time updating conditions, compared against the actual values. A comparison between Figure 6 and Figure 4, which displays the offline prediction results, reveals a significant improvement; in the offline setting, the models generated predictions based solely on the initial training data, and deviations were clear, particularly in the later cycles. However, under real-time updating conditions, the progressive retraining of the models with newly obtained operational data has allowed all five architectures to track the actual capacity curve more closely, with the prediction error in the final cycles being noticeably reduced.

Figures 7 to 9 present the MAE, RMSE, and R^2 values, respectively, for different values of K across all five architectures. Based on these results, the Conformer consistently demonstrates the best and most stable performance across all values of K ; its MAE ranges from 0.039 to 0.057 with a minimum at $K = 3$, RMSE lies within 0.052 to 0.069, and R^2 remains above 0.92 throughout, reaching a maximum of 0.956. The overall best performance of this architecture is achieved at $K = 3$. The stability of these metrics across different values of K indicates the robustness of the Conformer architecture against variations in the partitioning of update data. The Transformer-Encoder ranks second, with MAE ranging from 0.044 to 0.06, RMSE between 0.059 and 0.078, and R^2 predominantly above 0.90 with a peak of 0.941. The best performance of this model is obtained at $K = 10$. The acceptable stability of this model, combined with its lower memory and

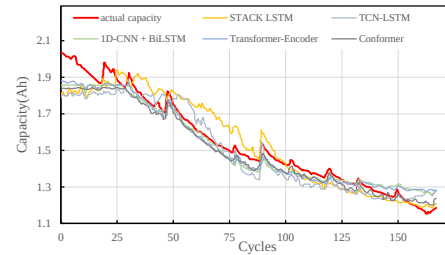
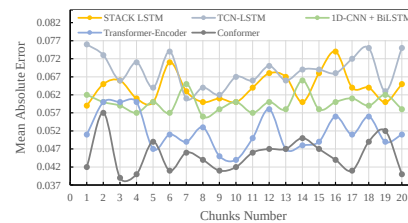
Figure 6: Real-Time Capacity Prediction Results of Deep Learning Models at Optimal K Values

Figure 7: Sensitivity of MAE to the Number of Real-Time Data Chunks

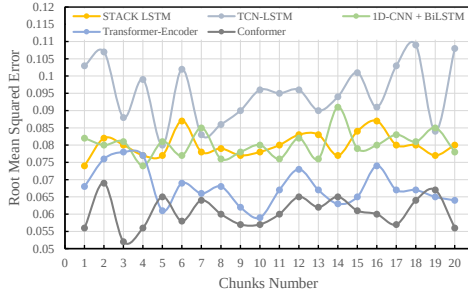
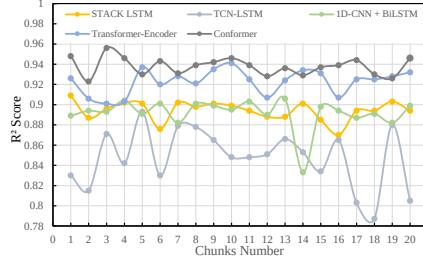


Figure 8: Sensitivity of RMSE to the Number of Real-Time Data Chunks

Figure 9: Sensitivity of R^2 to the Number of Real-time Data Chunks

processing time requirements compared to the Conformer, makes it a suitable alternative in computationally constrained scenarios. Stacked LSTM and 1D-CNN-BiLSTM exhibit intermediate performance. Stacked LSTM records MAE values between 0.059 and 0.074, RMSE between 0.074 and 0.087, and R^2 between 0.876 and 0.909, with its best performance observed at $K=1$. 1D-CNN-BiLSTM also yields MAE between 0.056 and 0.066, RMSE between 0.074 and 0.091, and R^2 between 0.833 and 0.904, with its best result obtained at $K=4$. Both models remain stable at lower values of K , yet as K increases beyond 15, their performance fluctuations become more pronounced, reflecting the sensitivity of recurrent structures to an increasing number of update stages and a decreasing volume of data per stage.

The weakest performance belongs to TCN-LSTM, with MAE between 0.061 and 0.076, RMSE between 0.083 and 0.109, and R^2 dropping to as low as 0.787 at certain values of K . The best performance of this model is achieved at $K=5$. The range of fluctuations in this model is also considerably wider than in the other architectures, which stems from the inefficient interaction between the TCN and LSTM layers under conditions where the volume of each update segment is reduced. Overall, an excessive increase in K not only fails to yield improvement

but also introduces instability in some architectures. Attention-based architectures, including the Conformer and Transformer-Encoder, maintain consistent performance across a broad range of K , and the choice of K becomes more relevant from a computational cost perspective. In contrast, for recurrent architectures, intermediate values of K in the range of approximately 3 to 7 establish the best tradeoff between accuracy and stability.

4.3. Impact of Ensemble Aggregation Strategies

This section investigates the impact of ensemble aggregation strategy selection on real-time battery capacity prediction accuracy within the digital twin framework based on cumulative incremental learning. In the preceding section, the performance characteristics of each base architecture were assessed independently; the focus now shifts to how combining the outputs of these models through six different aggregation strategies, namely SAE, IEWE, NMOE, ERWE, TME, and ACWE, can improve prediction accuracy and stability. All strategies were evaluated over chunk sizes ranging from 1 to 20, and the results are reported based on three metrics: MAE, RMSE, and R^2 . Figures 10 to 12 present the values of the three-evaluation metrics for all six strategies over chunk sizes ranging from $K=1$ to $K=20$. Across all values of K and all three metrics, NMOE and ERWE consistently achieve the best performance among the six strategies. This sustained superiority can be attributed to the weight-computation mechanisms employed by each: NMOE directly optimizes the weight vector to minimize the combination error on validation data. At the same time, ERWE assigns weights based on the ordinal ranking of models rather than absolute error values. Among the remaining strategies, ACWE exhibits the weakest performance; this strategy computes the weight of each model separately for each sample based on local confidence and overall reliability, yet in the incremental learning setting where the data volume per chunk is limited, such sample-level weighting leads to overfitting rather than improved accuracy, thereby degrading its overall performance.

Based on the sensitivity analysis conducted on the six aggregation strategies over chunk sizes from $K=1$ to $K=20$, $K=10$ is identified as the optimal chunk size. At this point, NMOE achieves the highest accuracy among the six ensemble strategies with MAE=0.0319, RMSE=0.0481, and $R^2=0.963$, while ERWE ranks second with MAE=0.0375, RMSE=0.0555, and $R^2=0.9513$. Figure 13 illustrates the predicted capacity trajectory of all six strategies compared to the actual values; as can be observed, the NMOE curve shows the closest agreement with the actual capacity degradation trend, particularly

TABLE3. Capacity estimation performance of individual models and ensemble strategies at offline and real-time updating levels

APPROACHES	Mean Absolute Error			Root Mean Squared Error			R^2 Score		
	Battery DT without updating	Battery DT with Data Real-time updating	Improvement	Battery DT without updating	Battery DT with Data Real-time updating	Improvement	Battery DT without updating	Battery DT with Data Real-time updating	Improvement
Stacked LSTM	0.085	0.059	30.59%	0.098	0.074	24.49%	0.846	0.909	7.45%
TCN-LSTM	0.092	0.064	30.43%	0.118	0.080	32.20%	0.779	0.893	14.63%
1D-CNN-BiLSTM	0.098	0.057	41.84%	0.119	0.074	37.82%	0.773	0.904	16.95%
Transformer Encoder	0.062	0.044	29.03%	0.089	0.059	33.71%	0.874	0.941	7.67%
Conformer	0.052	0.039	25.00%	0.070	0.052	25.71%	0.922	0.956	3.69%
SAE	0.057	0.042	26.32%	0.076	0.061	19.74%	0.907	0.940	3.64%
IEWE	0.056	0.040	28.57%	0.075	0.058	22.67%	0.909	0.945	3.96%
NMOE	0.048	0.031	35.42%	0.066	0.048	27.27%	0.929	0.963	3.66%

ERWE	0.053	0.0375	29.25%	0.067	0.055	17.91%	0.927	0.951	2.59%
TME	0.060	0.043	28.33%	0.082	0.062	24.39%	0.892	0.937	5.04%
ACWE	0.060	0.043	28.33%	0.083	0.064	22.89%	0.889	0.934	5.06%

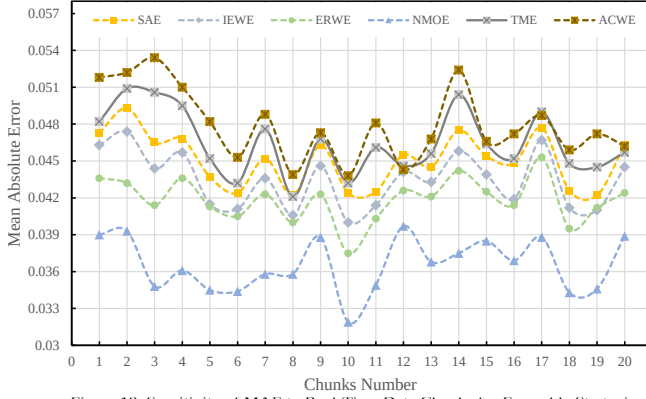


Figure 10: Sensitivity of MAE to Real-Time Data Chunks for Ensemble Strategies

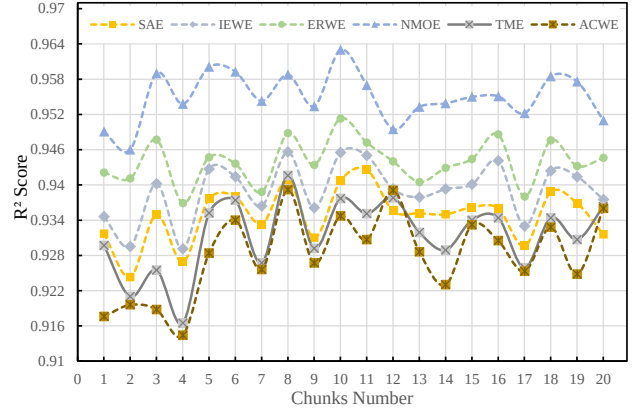
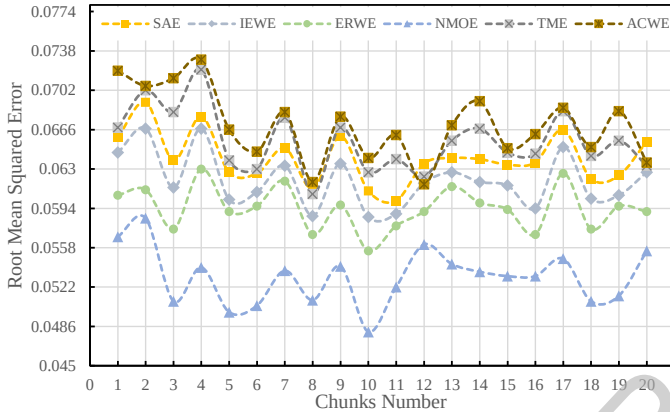
Figure 12: Sensitivity of R^2 to Real-time Data Chunks for Ensemble Strategies

Figure 11: Sensitivity of RMSE to Real-Time Data Chunks for Ensemble Strategies

during the mid-life and end-of-life phases of the battery, where accurate prediction becomes more critical. Figure 14 summarizes the values of the three evaluation metrics for all strategies and confirms the superiority of NMOE and ERWE. Furthermore, the results indicate that the ensemble aggregation approach has overall improved prediction accuracy compared to the individual base models; that is, combining the outputs of models with different architectures and algorithms has compensated for a considerable portion of individual errors and yielded a more stable final prediction.

4.4. Overall Comparison and Discussion

Capacity prediction accuracy within the proposed digital twin framework was evaluated across three successive levels. At the first level, the base models serving as virtual twins of the battery were trained offline on historical data. At the second level, by incorporating the cumulative incremental learning mechanism, the models were adapted in real time using operational data received from the physical battery. At the third level, ensemble strategies combined the outputs of the five updated models. The detailed analysis presented in this section focuses on the NASA dataset. Table 3 presents the performance of all approaches under both offline and real-time update scenarios, along with the percentage improvement for each metric. At the offline level, NMOE outperforms all individual models and other ensembles across all three metrics of MAE, RMSE, and R^2 . However, its margin over Conformer, the best-performing individual model, remains limited. The MAE values for NMOE and Conformer are reported as 0.048 and 0.052, respectively, and SAE with an MAE of 0.057 also performs close to Conformer. This proximity indicates that in the absence of real-time updating, a powerful single architecture can compete with many intermediate ensembles, and merely aggregating multiple models does not necessarily lead to superiority. On the other hand, all six ensemble methods maintain a considerable

TABLE4. Remaining useful life prediction results for individual models and ensemble strategies at offline and real-time updating levels

APPROACHES	Battery DT without updating				Battery DT with Data Real-time updating			
	Rul rea	Rul pre	Rul ae	Rul ape	Rul rea	Rul pre	Rul ae	Rul ape
Stacked LSTM	53	45	8	15%	53	49	4	7.5%
TCN-LSTM	53	56	3	5.6%	53	52	1	1.8%
1D-CNN-BiLSTM	53	48	5	9.4%	53	51	2	3.7%
Transformer Encoder	53	55	2	3.7%	53	54	1	1.8%
Conformer	53	47	6	11.3%	53	52	1	1.8%
Simple Average Ensemble (SAE)	53	48	5	9.4%	53	52	1	1.8%
Inverse Error-Weighted Ensemble (IEWE)	53	49	4	7.5%	53	52	1	1.8%
Nelder-Mead Optimized Ensemble (NMOE)	53	52	1	1.8%	53	53	0	0%
Exponential Rank-Weighted Ensemble (ERWE)	53	56	3	5.6%	53	53	0	0%
Trimmed Mean Ensemble (TME)	53	47	6	11.3%	53	51	2	3.7%

Adaptive Confidence-Weighted Ensemble (ACWE)	53	45	8	15%	53	56	3	5.6%
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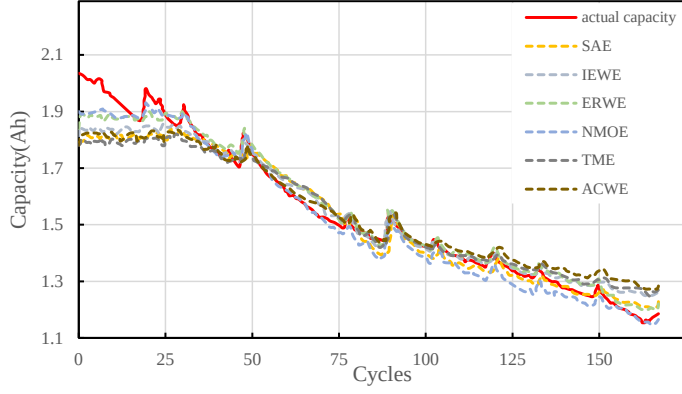
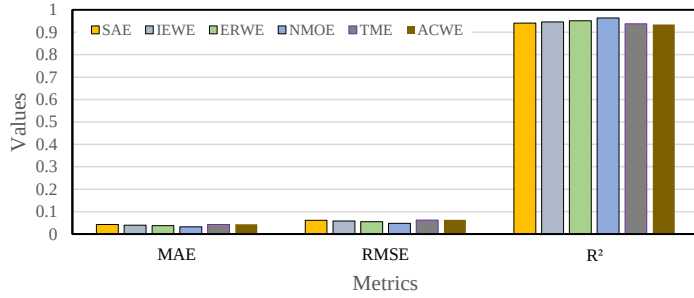
Figure 13: Real-time Capacity Prediction Results of Ensemble Strategies at Optimal K Values

Figure 14: Performance Comparison of Ensemble Strategies at Optimal Chunk Size

gap over the less accurate individual models. For instance, TME and ACWE, which have the lowest accuracy among the ensembles with an MAE of 0.060, still perform significantly better than 1D-CNN-BiLSTM with an MAE of 0.098. Overall, these results suggest that at the offline level, the performance gap between intermediate ensembles and strong individual models is not substantial, and the choice of aggregation mechanism is as decisive as the choice of the base model architectures themselves.

Upon applying incremental learning, this situation changes, and the distinction among approaches becomes more pronounced. The MAE gap between NMOE and Conformer, which was 0.004 at the offline level, widens to 0.008 after updating, with MAE values reaching 0.031 and 0.039, respectively. Moreover, NMOE achieves an R^2 of 0.963, the highest value across the entire table. In contrast, the updated Conformer with an MAE of 0.039 outperforms both TME and ACWE, each having an MAE of 0.043, and even surpasses SAE with an MAE of 0.042. At the bottom of the table, 1D-CNN-BiLSTM with an MAE of 0.057 and TCN-LSTM with an MAE of 0.064 still lag noticeably behind even the weakest ensembles after updating. Consequently, the proposed three-level evaluation protocol demonstrates that each component of the digital twin framework, namely the base models, real-time incremental learning, and ensemble strategies, makes an independent and measurable contribution to improving prediction accuracy. The final performance of the integrated framework at the third evaluation level exhibits a gradual and consistent improvement over the preceding levels.

Table 4 presents the remaining useful life prediction results. In this evaluation, the prediction starting point is the cycle at which the actual battery capacity reaches 1.7 Ah, and the end-of-life threshold is defined as 70% of the nominal capacity, i.e., 1.4 Ah; accordingly, the true remaining useful life is obtained as 53 cycles. At the offline level, the absolute error RUL_{ae} ranges from 1 to 8 cycles, but with the application of incremental learning, this range narrows to 0 to 4 cycles. After updating, NMOE and ERWE achieve a zero-cycle error with $RUL_{ape} = 0\%$, whereas Conformer as the best individual model still exhibits a one-cycle deviation. The ranking of approaches in RUL prediction is fully consistent with their ranking in the continuous capacity estimation metrics, including MAE, RMSE, and R^2 ,

TABLE5. Absolute RUL prediction error across different methods and datasets

Ref	Prediction method	NASA	oxford
[19]	KF-EM-RTS	6	-
[38]	BiGRU-TCMPPO (DTV)	1	354
	BiGRU-TCMPPO (DTC)	2	212
	BiGRU-TCMPPO (CCCT)	1	242
	BiGRU-TCMPPO (Fusion)	0	19
[61]	MOGPR	8	226
[62]	LSTM	5	196
[63]	IWOA-PF	1	100
Proposed framework	Conformer	1	100

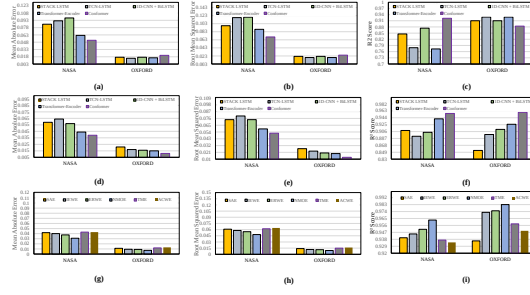


Figure 15: Comparative evaluation on NASA and Oxford datasets: (a–c) models with offline learning, (d–f) models with real-time updating, (g–i) models with ensemble strategy

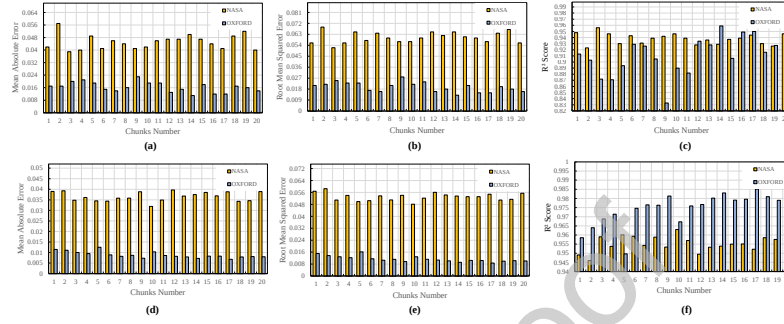


Figure 16: Sensitivity analysis of the number of chunks on prediction performance for the NASA and Oxford datasets: (a–c) single model, (d–f) ensemble model

confirming that higher accuracy in estimating the capacity degradation curve translates into higher accuracy in determining the end-of-life point. These results demonstrate that the proposed integrated digital twin framework, through the simultaneous integration of diverse deep learning architectures, cumulative incremental learning, and ensemble strategy, is capable of meeting the high accuracy requirements of real-time battery prognostic applications. To further validate the robustness and generalizability of the proposed framework beyond the NASA dataset, the same three-level evaluation protocol was applied to the Oxford battery dataset. Figure 15 presents a comparative evaluation across both datasets for individual models with offline learning (a–c), individual models with real-time updating (d–f), and ensemble strategies (g–i). The results reveal that while absolute error magnitudes differ between datasets due to their distinct operating conditions, the relative ranking of methods remains consistent. Specifically, NMOE maintains its superiority across both datasets, achieving the lowest MAE and highest R^2 in all three evaluation levels. On the Oxford dataset, NMOE attains an MAE of 0.007 after real-time updating, compared to 0.009 for the next-best ensemble method. Similarly, individual models exhibit the same performance hierarchy observed on NASA data: Conformer consistently outperforms other single architectures. This cross-dataset consistency confirms that the proposed framework demonstrates robust and generalizable performance across diverse battery degradation scenarios, validating its effectiveness for practical deployment in different operational environments.

Figure 16 reports a sensitivity analysis of the number of incremental chunks on prediction performance for both datasets, at the individual model level (a–c) and at the ensemble level (d–f). For the NASA dataset, individual models achieve optimal performance at approximately 3 chunks, with MAE stabilizing at 0.039. The ensemble approach on NASA data reaches its best accuracy at 10 chunks with an MAE of 0.031. Similarly, on the Oxford dataset, both individual models and the ensemble converge at 14 chunks, achieving MAEs of 0.011 and 0.007, respectively. These results indicate that performance plateaus after a limited number of chunks for both datasets, suggesting that the framework achieves reliable predictions without requiring the entire operational history. This behavior confirms that the incremental learning mechanism efficiently extracts relevant degradation patterns

from recent observations, enabling the digital twin to maintain high fidelity with minimal computational overhead. The consistency of saturation points across both datasets reinforces the robustness of the proposed framework and makes it suitable for edge-computing implementations where memory and processing power are limited. Finally, Table 5 compares the absolute RUL prediction error of the proposed approach against several existing methods on both NASA and Oxford datasets, confirming its superior accuracy across diverse battery degradation scenarios.

5. Conclusion

This study presented an intelligent digital twin framework for capacity estimation and RUL prediction of lithium-ion batteries, integrating five deep learning architectures, a cumulative incremental learning strategy, and six ensemble aggregation methods. It was validated through a three-level evaluation protocol on the NASA dataset, and its generalizability was further assessed on the Oxford dataset under diverse operational conditions. The primary contribution of this work lies in the design of a real-time update mechanism based on cumulative incremental learning, a strategy in which the model is retrained on the entire accumulated history upon receiving each new chunk of operational data, thereby preserving previously acquired knowledge while simultaneously adapting to the evolving dynamics of the battery degradation process. On the NASA dataset, this strategy improved the performance of all architectures compared to their offline counterparts; Conformer, as the best-performing individual model, improved from $R^2 = 0.922$ in the offline setting to $R^2 = 0.956$, yet attention-based models exhibited remarkable stability across a wide range of chunk sizes. In contrast, recurrent models delivered satisfactory performance only at small to moderate values. Beyond the individual model level, aggregating the outputs of the updated models played a decisive role; the optimization-based ensemble strategies, NMOE with $R^2 = 0.963$ and ERWE with $R^2 = 0.951$, substantially outperformed the best individual model, and both achieved zero-cycle error in remaining useful life prediction. In contrast, fixed-weight ensembles performed even worse than the standalone Conformer. This contrast indicates that merely combining diverse models is insufficient; the true benefit of architectural diversity is realized only

when each model's contribution is weighted proportionally to its performance. The consistent performance of the proposed framework across both datasets confirms its robustness under diverse operating conditions. Future research directions include validation on more diverse datasets and the development of a digital twin framework capable of jointly estimating the state of charge, state of health, and remaining useful life of batteries.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: